



VICTORIA JUNIOR COLLEGE

JC 2 PRELIMINARY EXAMINATION 2008

H1 CHEMISTRY

8872/1

PAPER 1 Multiple Choice

Thursday
18/09/2008

50 minutes
30 marks

Additional materials:

Data Booklet
Optical Mark Sheet

INSTRUCTIONS TO CANDIDATES

Do not open this booklet until you are told to do so.

Write your *name*, *CT group* and *NRIC / FIN* on the answer sheet in the spaces provided.

There are **thirty** questions in this paper. Answer **all** questions. For each question, there are four possible answers, **A**, **B**, **C** and **D**. Choose the **one** you consider correct and record your choice in **soft pencil** on the separate answer sheet.

Read very carefully the instructions on the answer sheet.

INFORMATION FOR CANDIDATES

Each correct answer will score one mark. A mark will not be deducted for a wrong answer.

A Data Booklet is provided.

Any rough working should be done in this booklet.

Section A

For each question there are four possible answers, **A**, **B**, **C**, and **D**. Choose the **one** you consider to be correct.

1 Which ion contains an unpaired electron?

- A** N^- **B** O^{2+} **C** S^- **D** Cl^+

2 To determine the percentage of nitrogen present in a snack, 1.0 g of the snack was boiled with concentrated sulphuric acid to convert all the nitrogen into ammonium sulphate. The ammonium salt obtained was then boiled with excess aqueous sodium hydroxide to liberate the ammonia which was passed into 25.0 cm³ of 0.20 mol dm⁻³ HCl. The unreacted HCl required 20.0 cm³ of 0.10 mol dm⁻³ NaOH for complete neutralisation.

What is the percentage by mass of nitrogen in the snack?

- A** 2.8% **B** 4.2% **C** 7.0% **D** 8.4%

3 In an experiment, 50 cm³ of 0.1 mol dm⁻³ solution of a metallic salt reacted exactly with 25.0 cm³ of 0.1 mol dm⁻³ aqueous sodium sulphite. The half-equation for the oxidation of the sulphite ion is shown below.



If the original oxidation number of the metal in the salt was +3, what would be the new oxidation number of the metal?

- A** +1 **B** +2 **C** +3 **D** +4

4 Two elements, **X** and **Y**, have the following properties.

- **X** and **Y** form ionic compounds Na_2X and Na_2Y respectively.
- Element **Y** forms YF_6 molecule whereas **X** is not able to do so.

Which pair of electronic configurations of **X** and **Y** is correct?

- | | X | Y |
|----------|---------------------------------------|---------------------------------------|
| A | $[\text{He}] 2\text{s}^2 2\text{p}^2$ | $[\text{Ne}] 3\text{s}^2 3\text{p}^2$ |
| B | $[\text{He}] 2\text{s}^2 2\text{p}^2$ | $[\text{Ne}] 3\text{s}^2 3\text{p}^4$ |
| C | $[\text{He}] 2\text{s}^2 2\text{p}^4$ | $[\text{Ne}] 3\text{s}^2 3\text{p}^2$ |
| D | $[\text{He}] 2\text{s}^2 2\text{p}^4$ | $[\text{Ne}] 3\text{s}^2 3\text{p}^4$ |

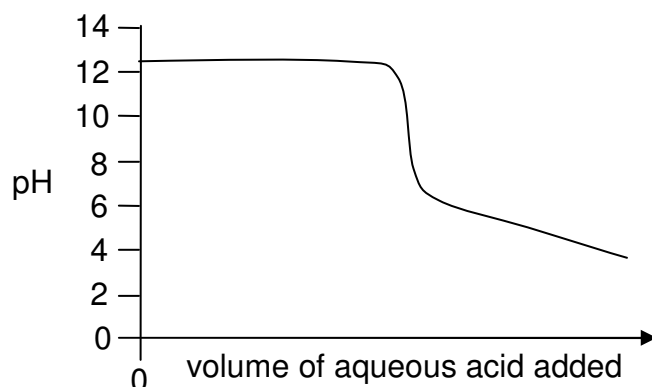
- 5 Which compound has a giant ionic lattice?
- A** Al_2O_3 **B** SiO_2 **C** Al_2Cl_6 **D** SiCl_4
- 6 The boiling points of pentane and 2,2-dimethylpropane are 36°C and 9°C respectively. Which statement best explains the difference in their boiling points?
- A** There are more electrons in pentane than in 2,2-dimethylpropane.
B Pentane is a polar molecule whereas 2,2-dimethylpropane is non-polar.
C The covalent bonds in pentane are stronger than those in 2,2-dimethylpropane.
D The dispersion forces in pentane are stronger than those in 2,2-dimethylpropane.
- 7 The standard enthalpy change of formation of but-1-ene, $\text{C}_4\text{H}_8(\text{g})$, and ethene, $\text{C}_2\text{H}_4(\text{g})$, are x and $y \text{ kJ mol}^{-1}$ respectively. Calculate the standard enthalpy change of the reaction, $2\text{C}_2\text{H}_4(\text{g}) \rightarrow \text{C}_4\text{H}_8(\text{g})$.
- A** $x - 2y$ **B** $2y - x$ **C** $\frac{1}{2}(x + y)$ **D** $\frac{1}{2}(y - x)$
- 8 Which compound has the greatest numerical value for its lattice energy?
- A** BaS **B** LiF **C** MgO **D** NaCl
- 9 Which of the following enthalpy changes is positive?
- A** $\text{Ag}^+(\text{g}) + \text{e}^- \rightarrow \text{Ag}(\text{g})$
B $\text{HCl}(\text{aq}) + \text{NaOH}(\text{aq}) \rightarrow \text{NaCl}(\text{aq}) + \text{H}_2\text{O}(\text{l})$
C $\text{CH}_4(\text{g}) + 2\text{O}_2(\text{g}) \rightarrow \text{CO}_2(\text{g}) + 2\text{H}_2\text{O}(\text{l})$
D $\text{Cl}_2(\text{g}) \rightarrow 2\text{Cl}(\text{g})$
- 10 What is the effect of a catalyst on the rate constants, k_1 for the forward reaction and k_{-1} for the reverse reaction, and on the equilibrium constant K , for a reversible reaction?
- | | k_1 | k_{-1} | K |
|----------|-----------|-----------|-----------|
| A | increases | decreases | no effect |
| B | increases | decreases | increases |
| C | increases | increases | no effect |
| D | no effect | no effect | increases |

- 11 For the reaction shown below, the value of the equilibrium constant, K_c , is 1.28 at 1000 K.



At equilibrium, the concentration of **C** is found to be $0.150 \text{ mol dm}^{-3}$. What is the equilibrium concentration of **A**, in mol dm^{-3} , in the reaction mixture?

- A** 0.018 **B** 0.117 **C** 0.133 **D** 0.300
- 12 Given the equilibrium, $\text{C}(\text{s}) + \text{H}_2\text{O}(\text{g}) \rightleftharpoons \text{CO}(\text{g}) + \text{H}_2(\text{g})$. What are the units of K_c ?
- A** $\text{mol}^2 \text{ dm}^{-6}$ **B** mol dm^{-3} **C** $\text{mol}^{-1} \text{ dm}^3$ **D** no unit
- 13 The graph shows the variation of pH during the titration of an aqueous alkali with an aqueous acid of similar concentration.



What could be the alkali and acid in this titration?

- | | alkali | acid |
|----------|--------------------------|---|
| A | $\text{NH}_3(\text{aq})$ | $\text{CH}_3\text{CO}_2\text{H}(\text{aq})$ |
| B | $\text{NH}_3(\text{aq})$ | $\text{HCl}(\text{aq})$ |
| C | $\text{NaOH}(\text{aq})$ | $\text{CH}_3\text{CO}_2\text{H}(\text{aq})$ |
| D | $\text{NaOH}(\text{aq})$ | $\text{HCl}(\text{aq})$ |
- 14 Consider the equilibrium, $\text{HSO}_4^-(\text{aq}) + \text{HPO}_4^{2-}(\text{aq}) \rightleftharpoons \text{SO}_4^{2-}(\text{aq}) + \text{H}_2\text{PO}_4^-(\text{aq})$.

Which of the following is a conjugate acid–base pair?

- | | base | conjugate acid |
|----------|---------------------|---------------------------|
| A | HPO_4^{2-} | H_2PO_4^- |
| B | HPO_4^{2-} | HSO_4^- |
| C | HSO_4^- | HPO_4^{2-} |
| D | HSO_4^- | SO_4^{2-} |

15 The table gives data for the reaction between **X** and **Y** at constant temperature.

experiment	initial [X] / mol dm ⁻³	initial [Y] / mol dm ⁻³	initial rate / mol dm ⁻³ s ⁻¹
1	0.3	0.2	4.0×10^{-4}
2	0.6	0.4	1.6×10^{-3}
3	0.6	0.8	6.4×10^{-3}

What is the rate equation for the reaction?

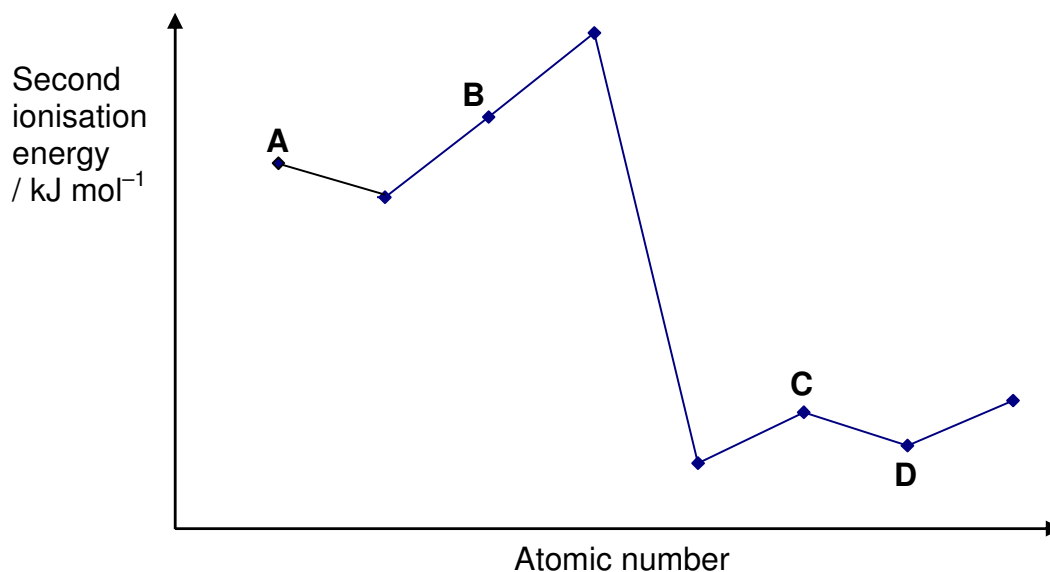
A rate = $k[\text{X}][\text{Y}]^2$

B rate = $k[\text{X}]^2[\text{Y}]$

C rate = $k[\text{X}]^2$

D rate = $k[\text{Y}]^2$

16 The following graph shows the second ionisation energy of eight consecutive elements in the Periodic Table.



Element **P** belongs to the third period of the Periodic Table. It has a high electrical conductivity and its chloride has a simple molecular structure.

Which point on the graph above does it correspond to the second ionisation energy of **P**?

17 Which statement does **not** describe the chemical properties of the compounds in the third period of the Periodic Table?

- A SiO_2 is formed from the hydrolysis of SiCl_4 .
- B Al_2O_3 is soluble in both acid and base.
- C Na_2O and P_4O_{10} can be mixed in water to give an approximately neutral solution.
- D The acidity of the chlorides in water decreases from NaCl to PCl_5 .

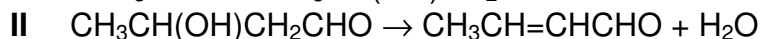
18 A student isolated an organic compound which decolourises aqueous bromine. The molecular formula of the organic compound is C_4H_8 . The total number of possible isomers, including *cis-trans* isomers, is

- A 3 B 4 C 5 D 6

19 The type of hybridisation of the carbon atoms in HCN , COCl_2 and CH_4 are

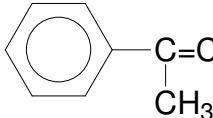
	HCN	COCl_2	CH_4
A	sp	sp^2	sp^3
B	sp	sp^3	sp^3
C	sp^2	sp^2	sp^2
D	sp^2	sp^3	sp^2

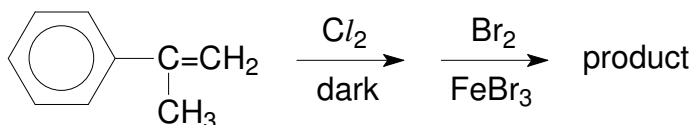
20 In the presence of a hydroxide catalyst, ethanal (CH_3CHO) can react with itself to form a compound known as aldol (reaction I). Aldol reacts to form another compound on heating (reaction II).



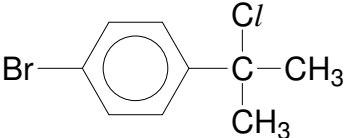
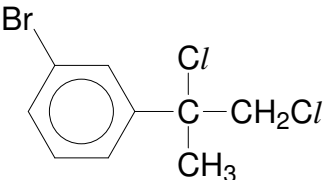
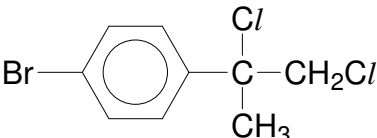
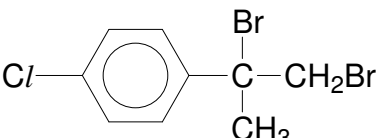
By examining the equations of the reactions or otherwise, choose the correct description of the reactions from the following.

	I	II
A	substitution	substitution
B	substitution	elimination
C	addition	substitution
D	addition	elimination

- 21  undergoes a series of reactions as shown.



What is the structural formula of the product formed?

- A** 
- B** 
- C** 
- D** 

- 22 An organic compound has the following properties. It forms a yellow precipitate with alkaline aqueous iodine. When warmed with acidified potassium dichromate(VI), the solution remains orange.

Which of the following could be the organic compound?

- A** $(\text{CH}_3)_2\text{CH}(\text{OH})$
B CH_3COCH_3
C $\text{CH}_3\text{CO}_2\text{CH}_3$
D $\text{CH}_3\text{CO}_2\text{H}$

23 Which of the following compounds **cannot** be synthesised from propan-1-ol within 2 steps or less?

- A ethanol
- B propan-2-ol
- C ethanoic acid
- D propanoic acid

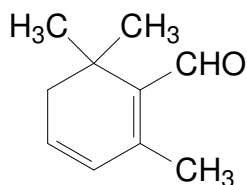
24 Compound **R** gives the following observations when reacted with the specific reagents.

- (I) Forms orange precipitate with 2,4-dinitrophenylhydrazine.
- (II) Liberates one mole of hydrogen gas with excess sodium.

Which of the following could be **R**?

- A $\text{OHCCH}_2\text{CH}(\text{OH})\text{CH}_3$
- B $\text{HO}_2\text{CCH}_2\text{COCH}_2\text{OH}$
- C $\text{HO}_2\text{C}-\text{C}_6\text{H}_4-\text{CH}_2\text{CH}(\text{OH})\text{CH}_3$
- D $\text{HO}_2\text{C}-\text{C}_6\text{H}_4-\text{CH}_2\text{COCH}_3$

25 Safranal is a component of the yellow dyestuff, saffron.



Safranal

What are the products of warming Safranal with Tollens' reagent?

- A a silver mirror and a carboxylate salt
- B a silver mirror and a carboxylic acid
- C a brick-red precipitate and a carboxylate salt
- D a brick-red precipitate and a carboxylic acid

Section B

For each of the questions in this section, one or more of the three numbered statements **1** to **3** may be correct.

Decide whether each of the statements is or is not correct (you may find it helpful to put a tick against the statements which you consider to be correct).

The responses **A** to **D** should be selected on the basis of

A	B	C	D
1, 2 and 3 are correct	1 and 2 only are correct	2 and 3 only are correct	1 only is correct

No other combination of statements is used as a correct response.

- 26** In an experiment, 10 cm^3 of an organic compound in the gaseous state was sparked with an excess of oxygen. 20 cm^3 of carbon dioxide and 5 cm^3 of nitrogen were obtained among the products. All gas volumes were measured at the same temperature and pressure.

Which molecular formulae would fit these data?

- 1** $\text{C}_2\text{H}_3\text{N}$
- 2** $\text{C}_2\text{H}_7\text{N}$
- 3** $\text{C}_2\text{H}_6\text{N}_2$

- 27** Which pairs of substances have similar types of bonding and structure?

- 1** BF_3 AlF_3
- 2** PH_3 SiCl_3H
- 3** Diamond SiC

- 28** Which statements explain the trend of the ionic radius across the third period of the Periodic Table?

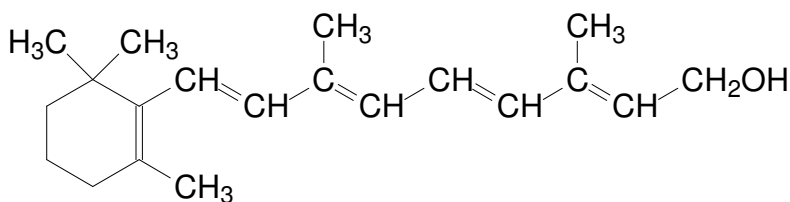
- 1** There is constant screening effect among the cations.
- 2** Nuclear charge increases across the period.
- 3** The anions have greater screening effect than the cations.

The responses **A** to **D** should be selected on the basis of

A	B	C	D
1, 2 and 3 are correct	1 and 2 only are correct	2 and 3 only are correct	1 only is correct

No other combination of statements is used as a correct response.

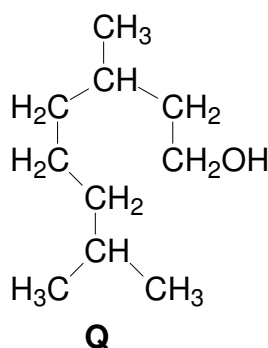
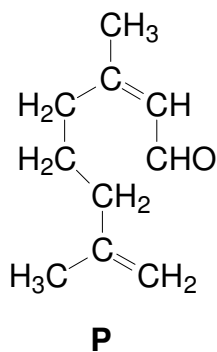
29 Vitamin A is an important nutrient that can prevent night-blindness.



Vitamin A

Which conclusions can be drawn from the structure of vitamin A?

- 1** It decolourises aqueous bromine.
 - 2** It reacts with 6 mol of HCl under suitable conditions.
 - 3** It has 32 geometric isomers.
- 30** The two compounds **P** and **Q** shown below are important flavours in citrus fruits. These compounds are commonly used in the food and perfume industries.



Which statements concerning **P** and **Q** are true?

- 1** **Q** can be distinguished from **P** by using phosphorus pentachloride.
- 2** Reaction of **P** with hot acidified KMnO_4 gives two organic products.
- 3** **P** can be converted to **Q** using lithium aluminium hydride in dry ether.